Properties of Linear Time-Invariant Systems:

$$\boxed{\begin{aligned} y[n] &= \sum_{k=-\infty}^{+\infty} x[k]\,h[n-k] = x[n] * h[n] \\ y(t) &= \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau = x(t) * h(t) \end{aligned}}$$

→ The unit impulse response does not completely characterize the behavior of a non-linear system.

e.g.
$$h[n] = \begin{cases} 1, & n = 0, 1 \\ 0, & \text{otherwise} \end{cases} \longrightarrow \text{(A)}$$

If the system is LTI then (A) characterizes all the i/o behavior.

e.g. $y[n] = x[n] + x[n-1]$

On the other hand, there are many nonlinear systems with the same response. e.g.

$$y[n] = \left(x[n] + x[n-1]\right)^2$$

$$y[n] = \max\left(x[n], x[n-1]\right).$$

— LTI systems have many properties that other systems do not posses.

The Commutative Property

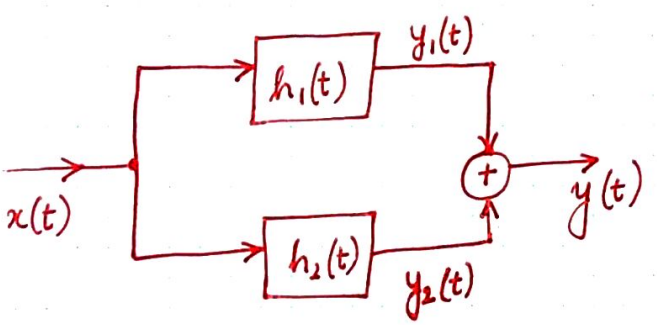
The output of an LTI system with input $x[n]$ and unit impulse response $h[n]$ is identical to the output of the LTI system with input $h[n]$ and unit impulse response $x[n]$.

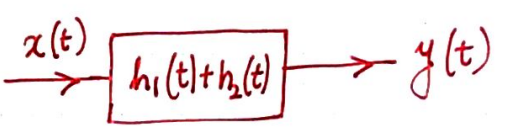
$$\boxed{\begin{aligned} x[n] * h[n] &= h[n] * x[n] \\ x(t) * h(t) &= h(t) * x(t) \end{aligned}}$$

The Distributive property

$$x[n] * \left(h_1[n] + h_2[n]\right) = x[n] * h_1[n] + x[n] * h_2[n]$$

$$x(t) * \left[h_1(t) + h_2(t)\right] = x(t) * h_1(t) + x(t) * h_2(t)$$





e.g. $x[n] = \left(\frac{1}{2}\right)^n u[n] + 2^n u[-n]$

$h[n] = u[n]$

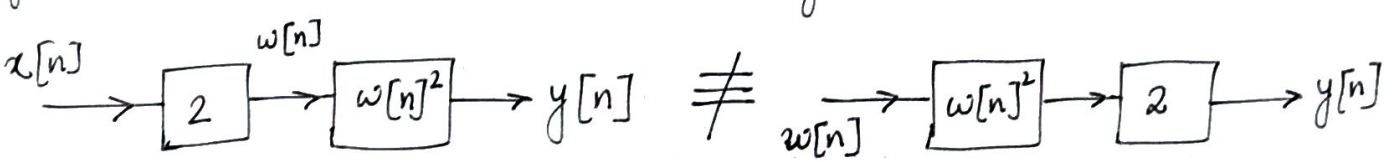
The Associative Property:

$$x[n] * \left(h_1[n] * h_2[n]\right) = \left(x[n] * h_1[n]\right) * h_2[n]$$

$$x(t) * \left(h_1(t) * h_2(t)\right) = \left(x(t) * h_1(t)\right) * h_2(t)$$

$\boxed{\square}$ Impulse response of the cascade of two LTI systems is the convolution of their individual impulse responses.

e.g. This is not true for nonlinear systems.



LTI systems without memory:

The only way a discrete time LTI system can be memoryless if $h[n] = 0$ for $n \neq 0$.

Therefore, the impulse has the following form:

$$h[n] = K\,\delta[n]$$

where $K = h[0]$ is a constant.

$$\therefore\ y[n] = \sum_{k=-\infty}^{\infty} x[k]\,K\,\delta[n-k] = K\,x[n]$$

For continuous-time: $y(t) = K\,x(t)$, $h(t) = K\,\delta(t)$

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k], \quad x(t) = \int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau)\,d\tau.$$

Memoryless LTI Systems:

The output of a discrete-time LTI system is:

$$y[n] = h[n] * x[n]$$

$$= \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$$

$$= \ldots\ldots + h[-2]\,x[n+2] + h[-1]\,x[n+1] + h[0]\,x[n] + h[1]\,x[n-1] + h[2]\,x[n-2] + \ldots\ldots \longrightarrow (A)$$

For $y[n]$ to be memoryless, $y[n]$ must depend only on $x[n]$ and therefore cannot depend on $x[n-k]$ for $k \neq 0$. Hence, every term in (A) has to be zero, except $h[0]x[n]$.

$$\therefore \quad h[k] = 0 \quad \text{for } k \neq 0.$$

A discrete LTI system is memoryless if and only if $h[k] = c\,\delta[k]$, where c is an arbitrary constant.

Writing the output of a continuous-time system as

$$y(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$$

Analogous to DT-LTI systems, a CT-LTI system is memoryless if and only if $h(\tau) = c\,\delta(\tau)$, for an arbitrary constant c.

All memoryless systems, simply perform scalar multiplication on the input.

Causal LTI Systems:

The output of a causal system depends only on past or present values of the input.

Again, we write the convolution sum as

$$y[n] = \ldots\ldots + h[-2]\,x[n+2] + h[-1]\,x[n+1] + h[0]\,x[n] + h[1]\,x[n-1] + \ldots\ldots$$

For the o/p to depend just on the past and present values we must have $h[k] = 0$ for $k < 0$.

$\therefore$ For DT causal LTI system,

$$h[k] = 0 \quad \text{for} \quad k < 0$$

and

$$y[n] = \sum_{k=0}^{\infty} x[k]\,h[n-k]$$

For continuous time - LTI system:

A causal continuous time - LTI system has an impulse response that satisfies:

$$h(\tau) = 0 \quad \text{for } \tau < 0.$$

The output is — $y(t) = \int_0^{\infty} h(\tau)\, x(t-\tau)\, d\tau.$

<u>Stable LTI system:</u>

Let us consider an input $|x[n]| < M_x < \infty$. The magnitude of the output is given by —

$$|y[n]| = |h[n] * x[n]|$$

$$= \left| \sum_{k=-\infty}^{\infty} h[k] \; x[n-k] \right|$$

$$\leqslant \sum_{k=-\infty}^{\infty} |h[k]|\,|x[n-k]|$$

$$\leqslant \sum_{k=-\infty}^{\infty} |h[k]|\, M_x$$

$$\leqslant M_x \sum_{k=-\infty}^{\infty} |h[k]|$$

$\therefore$ $|y[n]| < \infty$ provided $\sum_{k=-\infty}^{\infty} |h[k]| < \infty$.